

CERVICAL FASCIA

Superficial Cervical Fascia

It is a thin layer of fascia embedded in it the following structures:

- I. Platysma.
- II. Superficial veins.
- III. Cutaneous nerves.
- IV. Superficial cervical lymph nodes.

I. **Platysma**: It is a broad sheet of muscle fibers.

Origin:

Deep fascia covering deltoid and pectoralis major muscles.

Insertion:

- 1- Lower border of the body of mandible.
- 2- Blends with the muscles at the angle of the mouth.
- 3- The muscle fibers interlace with that of the opposite side below the symphysis menti.

Nerve supply: Cervical branch of facial nerve.

Action:

- 1- Wrinkles the skin of neck.
- 2- Depresses the angle of the mouth.
- 3- Assists in depression of the mandible.

II. **Superficial veins**: External jugular vein and its tributaries.

III. **Cutaneous nerves**:

They are the cutaneous branches of cervical plexus:

- 1- Lesser occipital nerve. 2- Great auricular nerve.
- 3- Transverse cutaneous nerve of neck. 4- Supraclavicular nerves.

IV. **Superficial cervical lymph nodes**:

Lie superficial to the sternomastoid muscle along the external jugular vein. They receive lymph from occipital and mastoid lymph nodes and drain it to the deep cervical lymph nodes.

The Deep Cervical Fascia

(The Deep Fascia of the Neck)

I. The Investing Layer of the Deep Fascia:

It surrounds the whole neck and splits to envelop, trapezius and sternomastoid muscles.

a- Posteriorly: It is continuous with ligamentum nuchae and C. 7 spine. Then, it splits to surround trapezius muscle and lastly it forms the roof of posterior triangle.

b- Laterally and anteriorly: The fascia splits again to surround the sternomastoid muscle then it covers the anterior triangle and becomes continuous with the corresponding fascia of the opposite side. In the middle, it is attached to:

(1) Symphysis menti. (2) Body of the hyoid bone. (3) Manubrium of sternum.

c- Superiorly: The fascia is attached to the superior nuchal line, mastoid process and lower border of mandible. Between the angle of mandible and the mastoid process, the fascia splits to cover the parotid gland forming the parotid fascia.

The superficial layer is fixed to the zygomatic arch and the deep layer to the styloid process and tympanic plate. The part of the parotid fascia extending from the styloid process to the angle of mandible is thickened forming the stylomandibular ligament.

d- Inferiorly: It is attached to the acromion, clavicle and manubrium. Short distance above the sternum, the fascia splits into two layers enclosing the suprasternal space which contains:

(1) Lower ends of anterior jugular veins. (2) The jugular arch connecting both anterior jugular veins.

(3) Sternal head of sternomastoid. (4) Some lymph nodes.

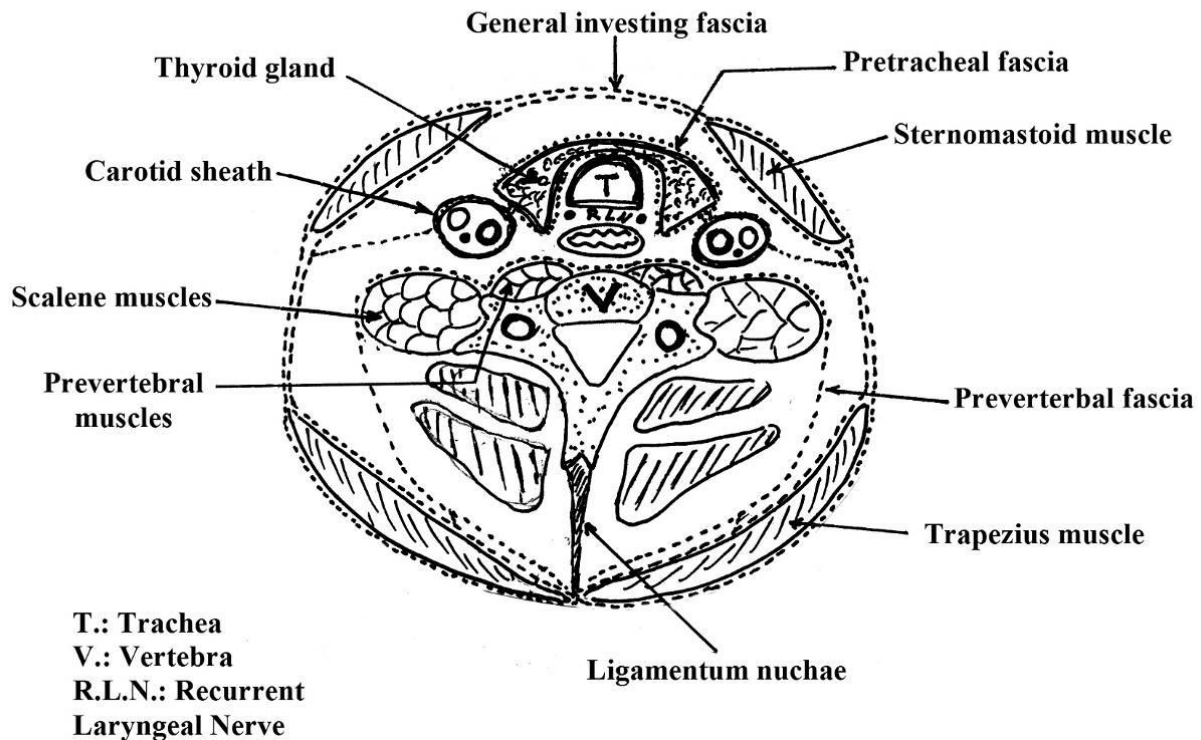
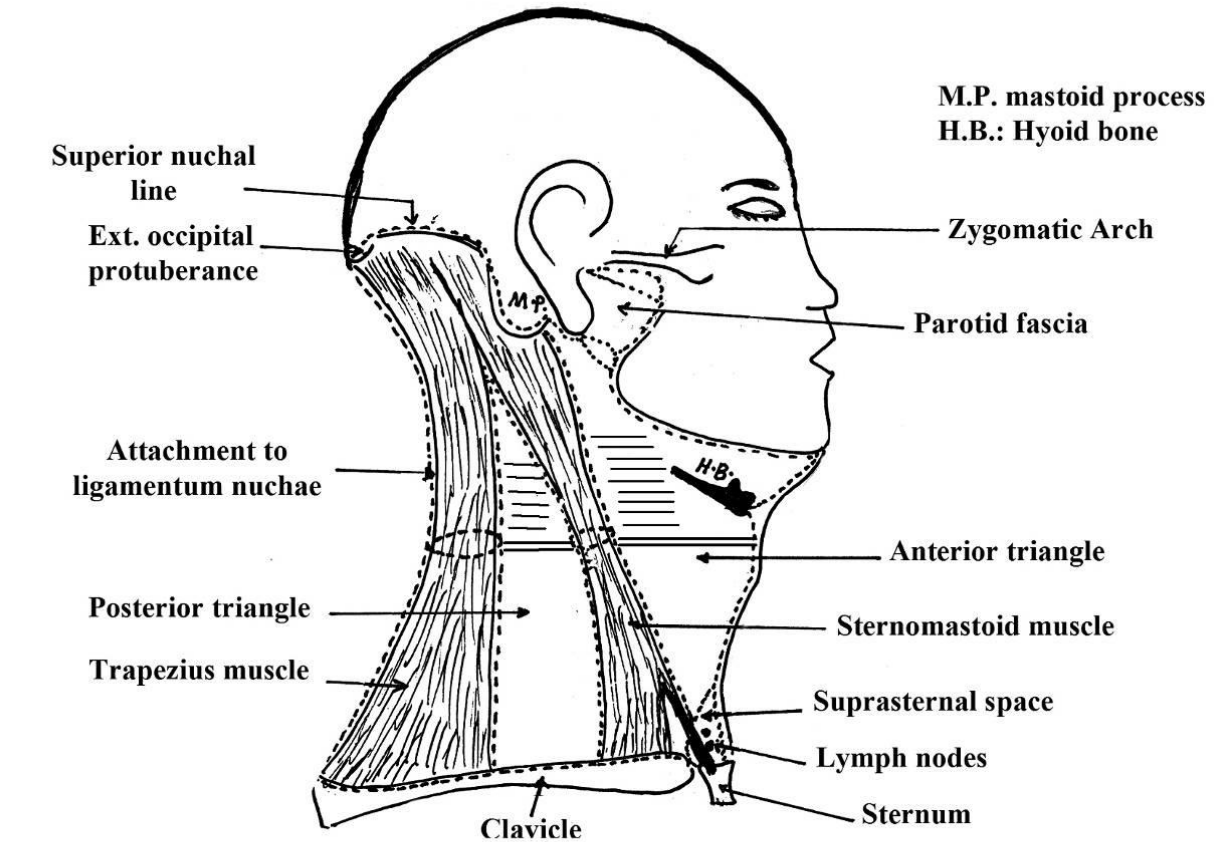
e- Structures that pierce the general investing fascia:

1- External jugular vein pierces the fascia one inch above the clavicle.

2- Anterior facial vein pierces it one finger below the mandible.

3- All cutaneous branches of cervical plexus.

Deep Fascia of the Neck



II. Prevertebral Fascia:

- It covers the prevertebral muscles and extends from the base of skull (above), to the third thoracic vertebra (below) where it blends with the anterior longitudinal ligament. Laterally, it covers the scalenus anterior muscle and the muscles

forming the floor of the posterior triangle (scalenus medius, levator scapulae and splenius capitis) then it fades under trapezius muscle.

- As the subclavian artery and the trunks of the brachial plexus emerge from beneath the lateral border of the scalenus anterior muscle, they take a tube of prevertebral fascia with them, behind the clavicle, which is called axillary fascia.

III. Pretracheal Fascia:

- A deep fascia lies in front and sides of trachea, it splits to invest the thyroid gland connecting it to the oblique line of thyroid and to the arch of cricoid cartilages (the larynx). That is why the thyroid gland moves up and down with larynx during swallowing.
- Laterally it blends with carotid sheath; inferiorly it invests the inferior thyroid veins and is continuous inside the thoracic cavity with the fibrous pericardium.

IV. Carotid Sheath:

- A tube of deep fascia that invests the big vessels and nerves of the neck except external carotid artery.
- It extends from the base of the skull till the root of the neck.

Attachments:

Above: It is attached to tympanic plate and petrous part of temporal bone.

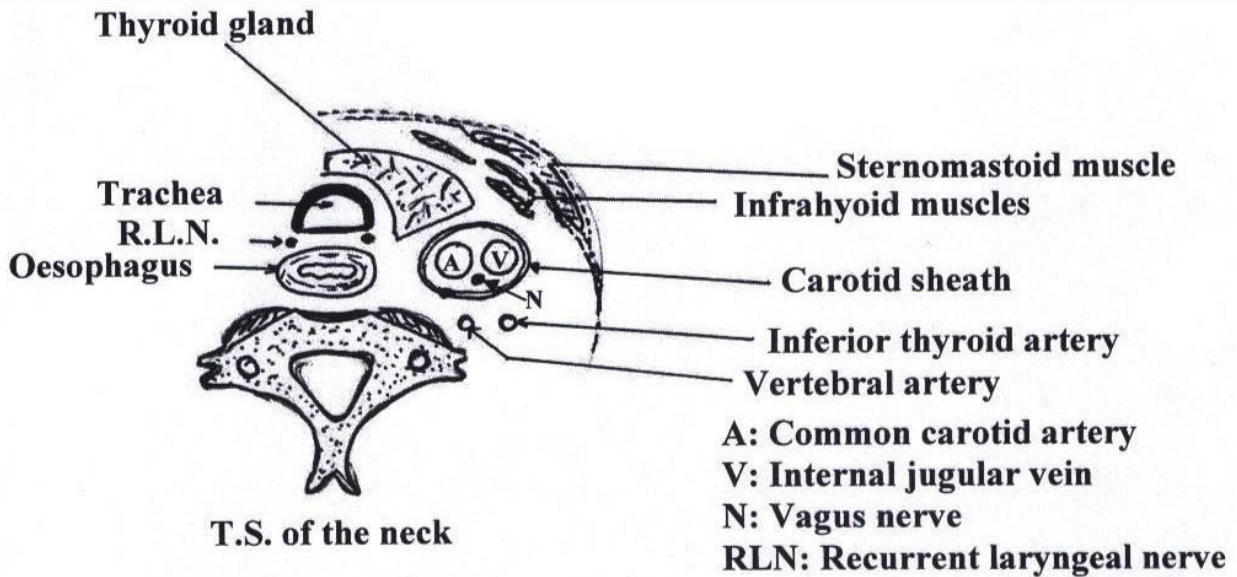
Anteriorly: It is connected to pretracheal fascia.

Posteriorly: It is continuous with prevertebral fascia.

Contents:

- 1- Common and internal carotid arteries medially.
- 2- Internal jugular vein laterally.
- 3- Vagus in between and slightly posteriorly.
- 4- Sympathetic chain is embedded in its posterior wall and ansa cervicalis is embedded in its anterior wall.

The upper part of the sheath, above the posterior belly of digastric muscle, contains the lower 4 cranial nerves while the lower part of the sheath contains the vagus nerve only.



Relations:

Anteriorly:

- 1- **Muscles:** Infrahyoid, sternomastoid.
- 2- **Gland:** Thyroid lobe.
- 3- **Veins:** Superior and middle thyroid.

Posteriorly:

- 1- Cervical vertebrae, prevertebral muscles and fascia.
- 2- Arteries: Inferior thyroid and vertebral arteries.

Medially:

- 1- Larynx and pharynx (above).
- 2- Trachea and oesophagus with recurrent laryngeal nerve in between (below).

V. Buccopharyngeal Fascia:

VI. Pharyngobasilar Fascia

External Jugular Vein

It **begins** near the angle of the mandible below the parotid gland by union of the posterior auricular vein and posterior division of retromandibular vein.

It descends obliquely on the surface of the sternomastoid, then it descends

superficial to the deep fascia of roof of the posterior triangle till about 1 inch above the clavicle where it pierces the deep fascia of the roof of the posterior triangle and continues downwards inside the triangle to end in the subclavian vein.

Tributaries:

- 1- Posterior auricular vein.
- 2- Posterior division of retromandibular vein.
- 3- Transverse cervical vein.
- 4- Suprascapular vein.
- 5- Anterior jugular vein.
- 6- Posterior external jugular vein: It drains the back of the scalp and upper part of the back of the neck and joins the middle part of the external jugular vein.

Surfaces anatomy of external jugular vein:

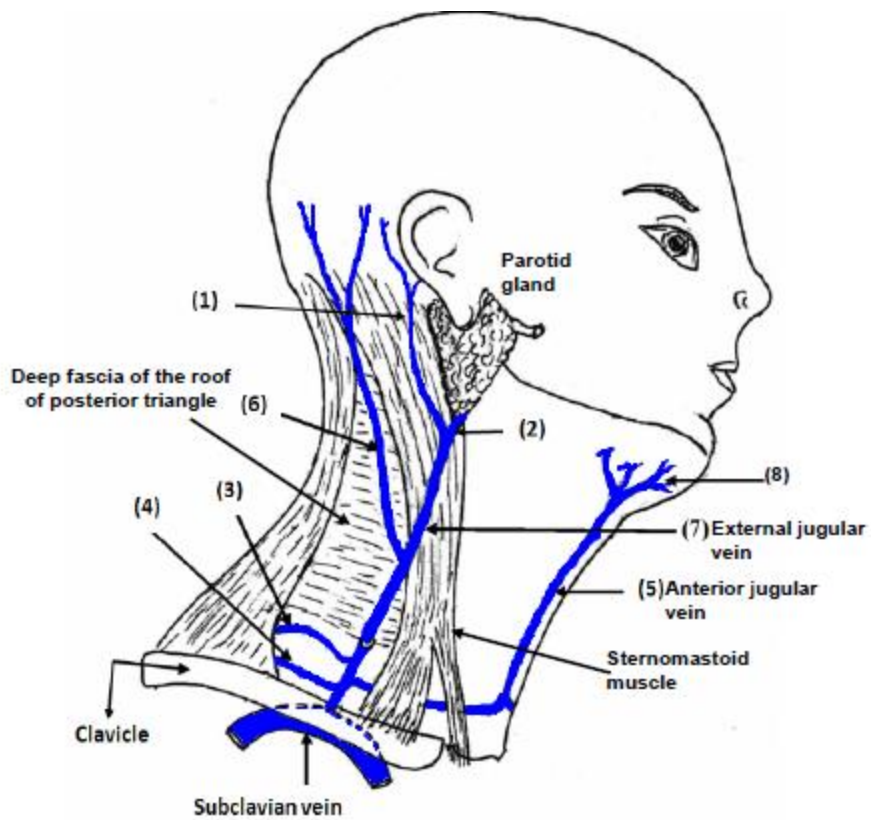
A line drawn from a point just below and behind the angle of mandible to a point at the middle of the clavicle.

Applied anatomy:

The external jugular vein is adherent to the deep fascia at the point where it pierces it, if the vein is injured at this point it will not collapse and air embolism occurs.

Anterior jugular vein:

It starts by union of submental veins just above the hyoid bone. It descends in the superficial fascia near the midline of neck to reach the suprasternal notch. The vein then shifts laterally in front of internal jugular vein and scalenus anterior muscle and deep to sternomastoid to end in external jugular vein. Above the sternum the two anterior jugular veins are connected together by jugular arch.



Veins in the Posterior Triangle of Neck

- (1) Posterior auricular vein
- (2) Posterior division of retromandibular vein
- (3) Transverse cervical vein
- (4) Suprascapular vein
- (5) Anterior jugular vein
- (6) Posterior external jugular vein
- (7) External jugular vein

Sternomastoid Muscle

It is an important landmark present between the anterior and posterior triangles.

Origin: By two heads:

1- Sternal head: Rounded, tendinous head which arises from the anterior surface of the upper part of manubrium of sternum.

2- Clavicular head: Broad, fleshy head which arises from the medial third of the upper surface of the clavicle.

Insertion: Lateral surface of mastoid process and lateral half of superior nuchal line.

Nerve supply:

1- Spinal part of accessory nerve (motor).

2- Ventral rami of C2 and C3 (sensory).

Action:

1- When one muscle acts alone, it bends the head to the same side and rotates the face to the opposite side.

2- When the two muscles act together they produce flexion of the neck by projecting the head forwards. They also raise the head from the supine position.

Relations:

1- **The spinal part of accessory nerve** pierces the muscle at the junction of the upper and middle thirds and runs within it to about its middle.

2- **Superficial relations:**

a- Skin, fascia with platysma.

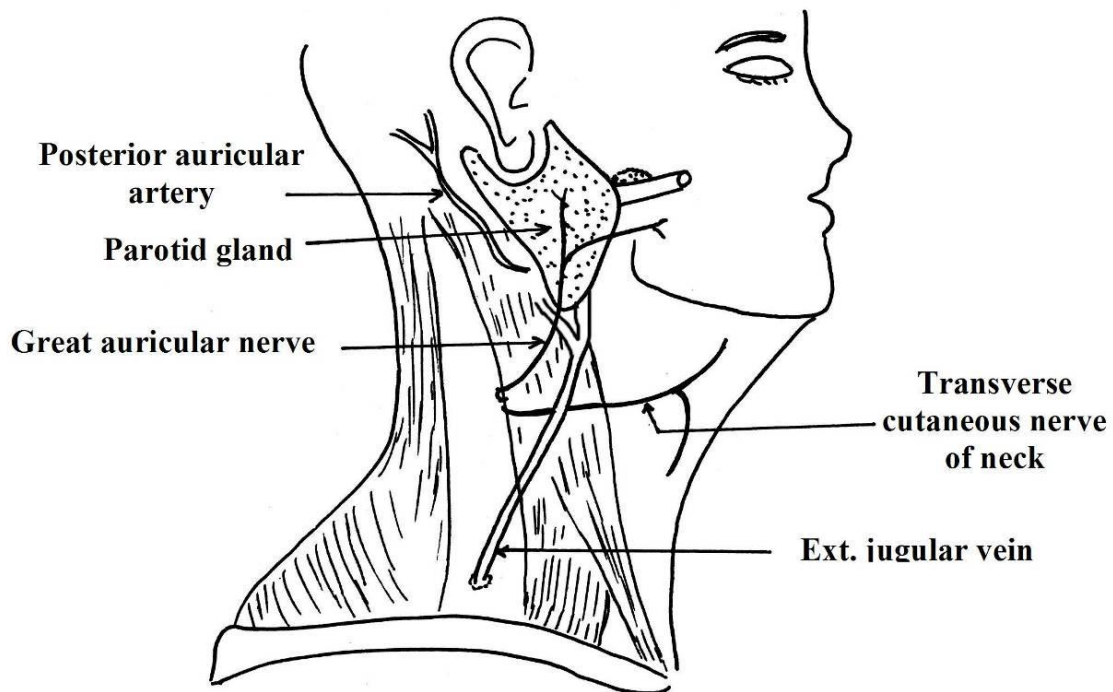
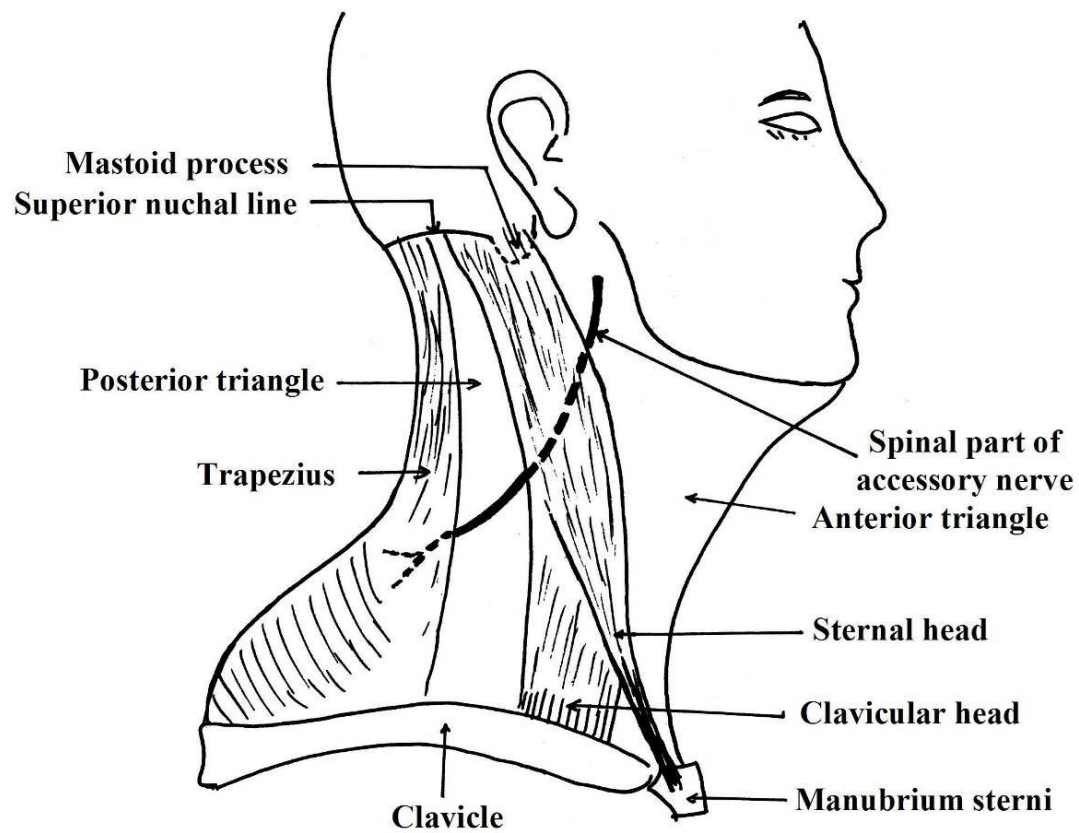
b- External jugular vein crosses it obliquely.

c- Great auricular nerve and transverse cutaneous nerve of neck.

d- Posterior auricular vessels.

e- Posteromedial surface of parotid gland.

Sternomastoid Muscle



Superficial Relations of Sternomastoid Muscle

3- Deep relations:

a- Carotid sheath and its contents: Two arteries (common and internal carotid), vagus nerve, and internal jugular vein.

b- Scalenus anterior muscle: With phrenic nerve, intermediate tendon of omohyoid, transverse cervical and suprascapular arteries on it.

c- Lobe of thyroid gland.

d- Eight muscles: Three infrahyoid muscles (sternohyoid, sternothyroid, intermediate tendon of omohyoid), 3 muscles in the floor of posterior triangle (scalenus medius, levator scapulae and splenius capitis), 2 other muscles:

scalenus anterior and posterior belly of digastric muscle.

e- Deep cervical lymph nodes, along the internal jugular vein.

f- Subclavian artery: (1st and 2nd parts).

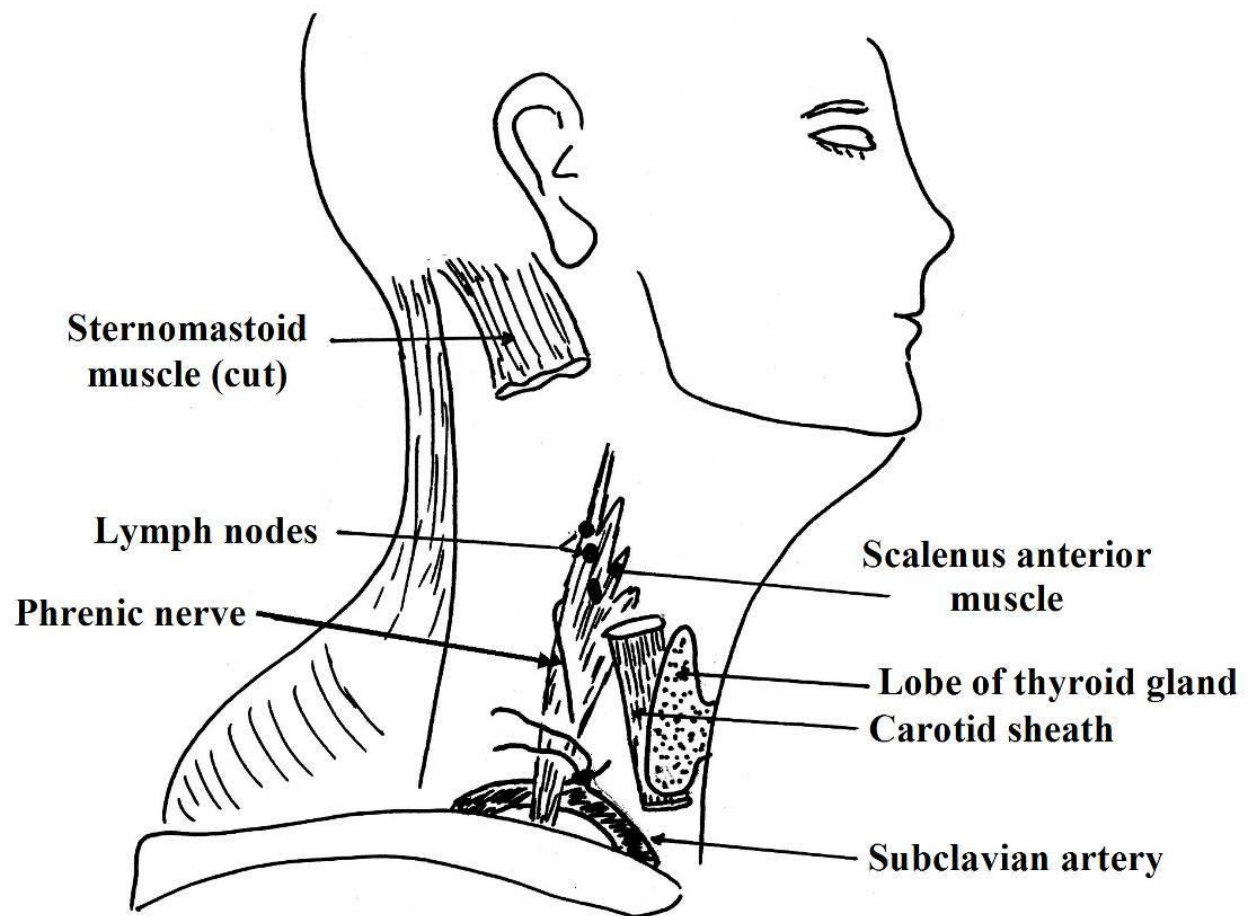
g- Mastoid process.

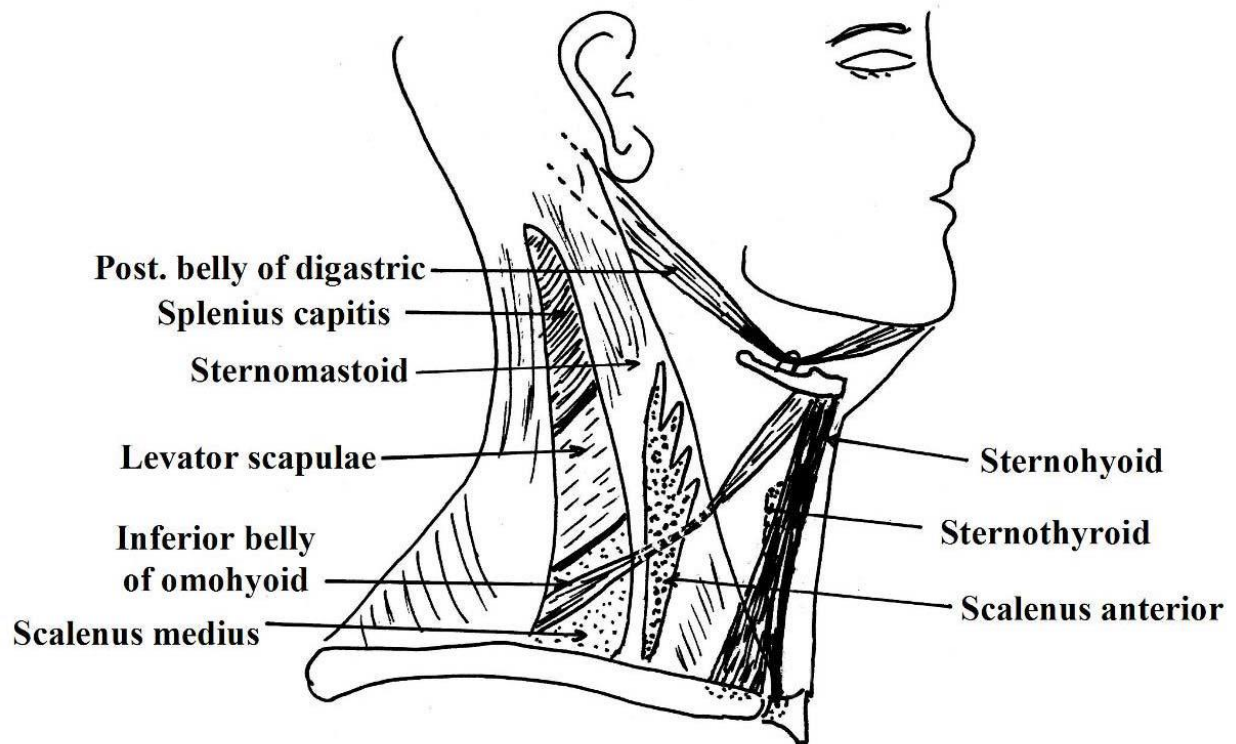
h- Anterior jugular vein in the lower part.

i- Thoracic duct on left side only.

j- Cervical plexus.

Deep Relations of Sternomastoid Muscle





Muscles Deep to Sternomastoid

Cervical Plexus

Position:

It lies opposite the upper 4 cervical vertebrae covered by the internal jugular vein and the sternomastoid muscle.

Formation:

It is formed by the union of the anterior rami of the upper 4 cervical nerves.

Branches:

A- Cutaneous branches

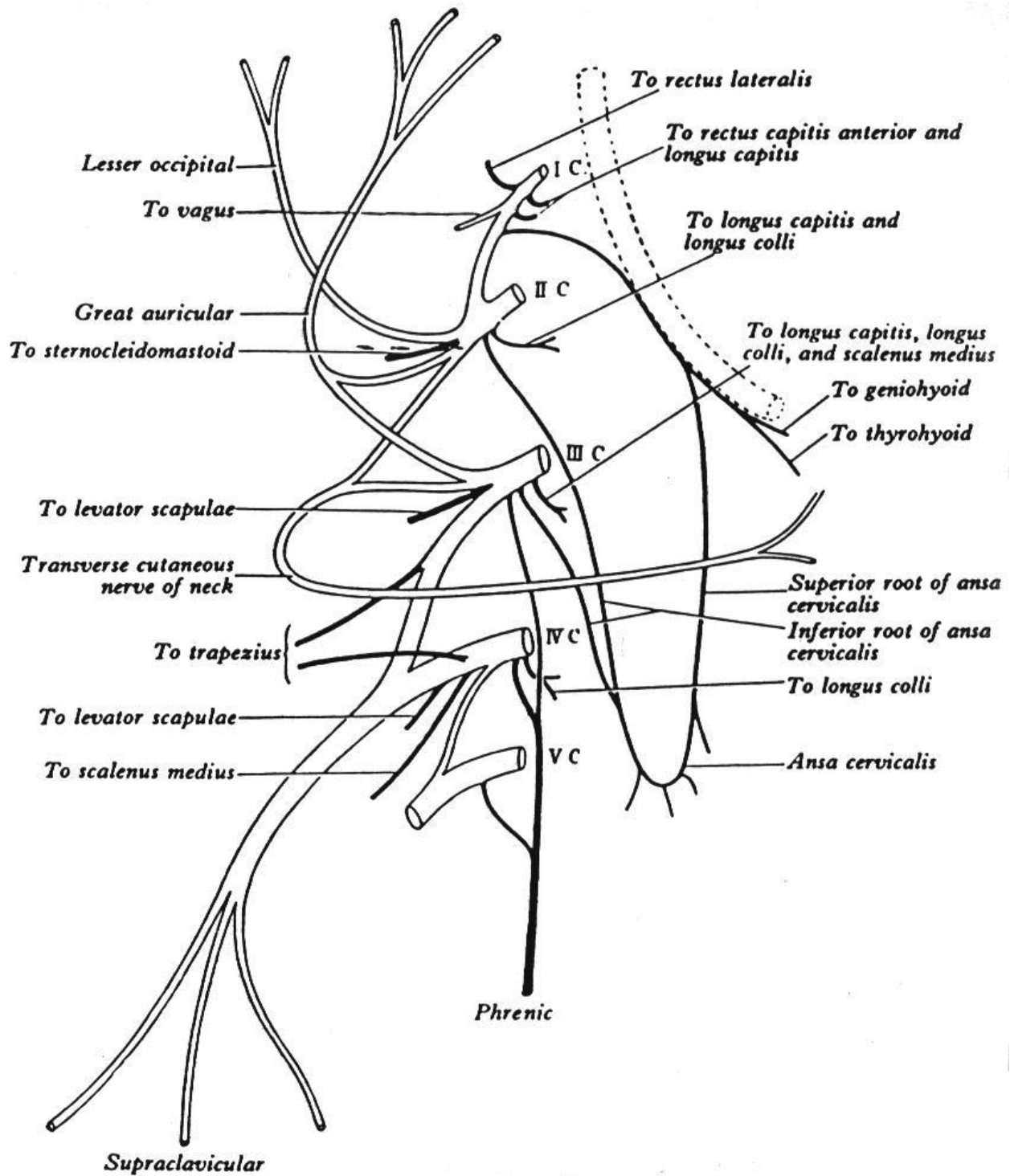
B- Muscular branches:

1- Descendens cervicalis (C2,3): It forms the inferior root of the ansa cervicalis and supplies the infrahyoid muscles except thyrohyoid muscle.

2- Phrenic nerve (C3,4,5): It descends on the surface of the scalenus anterior between the muscle and its fascia to supply the diaphragm.

3- Muscular branches: Lateral series to sternomastoid, trapezius, levator scapulae, scalenus medius. Medial series to rectus capitis anterior, rectus capitis lateralis, longus colli and longus capitis.

C- Communicating branches with the last 3 cranial nerves.



Subclavian Artery

Origin:

The right subclavian artery arises from the brachiocephalic artery behind the right sternoclavicular joint; while the left artery arises from the arch of aorta behind the manubrium sterni. Both arteries enter the neck behind the sternoclavicular joints.

Course:

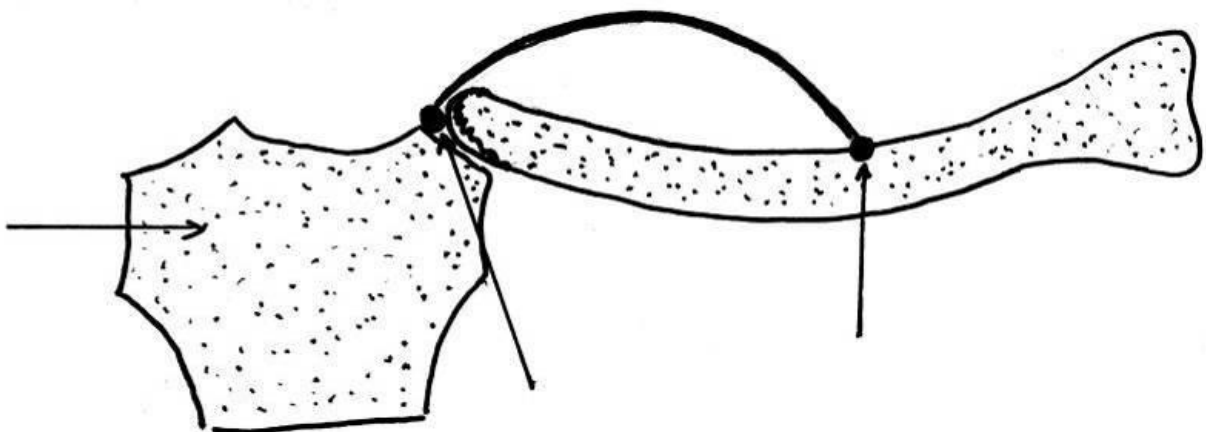
Each artery has an arched course in the root of the neck. It runs behind scalenus anterior muscle and in front of the apex of lung and cervical pleura.

Parts:

The scalenus anterior muscle divides the artery into 3 parts; 1st part (medial), 2nd part (deep), and 3rd part (lateral) to the muscle. Each artery ends at the outer border of the first rib to become the axillary artery.

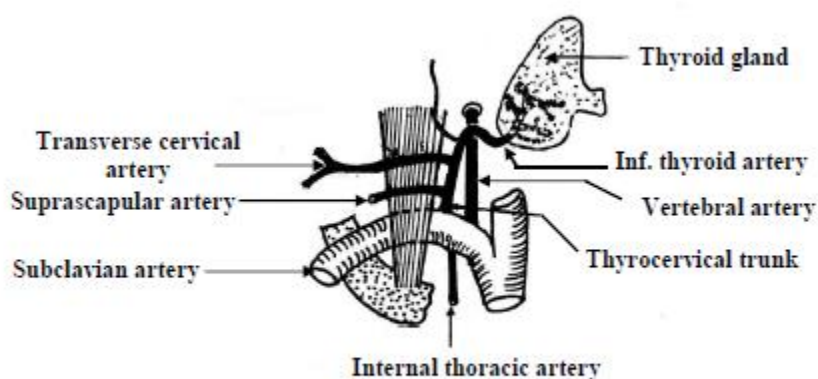
Surface anatomy of subclavian artery:

It is represented by a convex line, lying one inch above the clavicle and drawn from sternoclavicular joint to mid point of clavicle.



Relations and branches of subclavian artery:

	Anterior (in front)	Posterior (behind)	Branches
1st part (medial to scalenus anterior muscle)	1. Muscles: Sterno mastoid, sternohyoid, sternothyroid. 2. Nerves: vagus and its cardiac branches, sympathetic branches, ansa subclavia and phrenic nerve (on left side only). 3. Vein; internal jugular vein. 4. Thoracic duct (on left side only)	1. Apex of lung and cervical pleura 2. Suprapleural membrane. 3. Recurrent laryngeal nerve (on right side only)	1. Vertebral artery. 2. Internal thoracic artery. 3. Thyrocervical trunk: divides immediately into 3 branches: transverse cervical, suprascapular and inferior thyroid arteries
2nd part (deep to scalenus anterior muscle)	1. Sternomastoid 2. Scalenus anterior muscle with right phrenic nerve only.	As (1) and (2) of 1 st part	Costocervical trunk which divides into: deep cervical artery and superior intercostal arteries
3rd part (lateral to scalenus anterior muscle)	1. Skin and fascia of posterior triangle 2. Clavicle 3. Subclavian vein 4. External jugular vein	1. Lower trunk of brachial plexus. 2. Upper surface of 1st rib	Usually has no branches. It may give the descending scapular artery which replaces the deep branch of transverse cervical artery



Vertebral Artery

It **arises** from the upper and posterior aspects of 1st part of subclavian artery and is **divided** into 4 parts:

First part: It extends from subclavian artery till the foramen transversarium of the 6th cervical vertebra. It lies in the inverted V-shaped space medial to scalenus anterior muscle (vertebral triangle).

Second part: It ascends in foramina transversaria of upper 6 cervical vertebrae

surrounded by plexus of veins and sympathetic plexus of nerves.

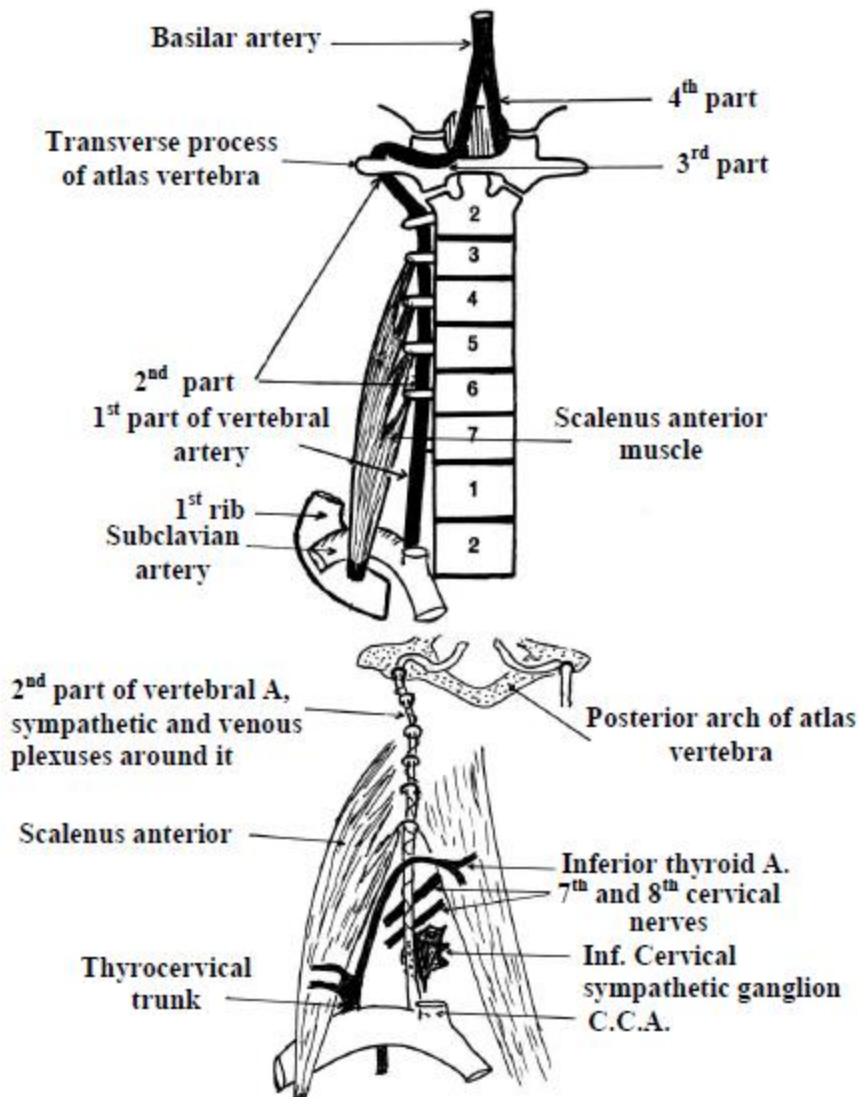
Third part: It lies in suboccipital triangle, grooving the upper surface of the posterior arch of atlas vertebra.

Fourth part: It enters the cranial cavity via foramen magnum. The 2 arteries unite together at the lower border of pons forming the basilar artery.

Branches:

- 1- First part has no branches.
- 2- Second part gives spinal branches to the spinal cord and its meninges.
- 3- Third part gives muscular branches to the surrounding muscles of the suboccipital triangle.
- 4- Fourth part gives anterior and posterior spinal arteries and posterior inferior cerebellar arteries.

Parts of Vertebral Artery



Relations of the Vertebral Artery

Thyrocervical Trunk

It is a short trunk which arises from the first part of subclavian artery close to the medial margin of the scalenus anterior muscle. It divides immediately into 3 branches:

1- Inferior thyroid artery.

2- Transverse cervical artery: It passes laterally, in front of the scalenus anterior muscle above the suprascapular artery. It then traverses the posterior triangle and divides into 2 branches (superficial

and deep) in relation to levator scapulae muscle. The deep branch descends along the medial border of scapula to share in

the anastomosis around the scapula. The superficial branch gives many branches to the deep surface of trapezius muscle.

3- Suprascapular artery: It passes laterally in front of scalenus anterior muscle then it crosses the lower part of posterior triangle, very close to the clavicle, and shares in the anastomosis around the scapula.

Costocervical Trunk

It is a short trunk which arises from the posterior surface of the 2nd part of subclavian artery. It passes backwards over the apex of the lung and divides into:

1- Deep cervical artery: Ascends between the muscles of the back of the neck near the midline to anastomose with the descending branch of the occipital artery and with the 3rd part of vertebral artery.

2- Superior intercostal artery: Descends in front of the neck of the 1st rib and ends by giving upper 2 posterior intercostal arteries.

Subclavian Vein

It **begins** as the continuation of the axillary vein at the outer border of the first rib.

Course:

runs on the upper surface of the 1st rib, in front of the scalenus anterior muscle

end it joins the internal jugular vein to form the brachiocephalic vein.

Tributaries: External jugular vein is the only tributary.

Central venous catheter:

A tested technique of quickly accessing the major venous system. Its benefits include rapid administration of combinations of drugs, a route for nutritional support and fluid administration.

Central venous pressure is considered a direct measurement about heart activity. It is acquired by putting a central venous catheter into any of several large veins either in IJV or subclavian vein.

Lateral Vertebral Muscles (Scalene Muscles)

Scalenus Anterior Muscle

Origin:

From the anterior tubercles of transverse processes of 3rd, 4th, 5th and 6th cervical vertebrae (the same origin of the rectus capitis anterior muscle).

Insertion:

Into the scalene tubercle and the adjoining ridge on the upper surface of the 1st rib.

Nerve supply: Lower cervical nerves (C4, 5, 6).

Action:

- 1- It bends the neck to the lateral side.
- 2- It helps in elevation of 1st rib in forced inspiration.

Relations:

A- Anterior relations:

Three muscles: Sternomastoid, subclavius, and intermediate tendon of omohyoid.

Three arteries:

- a- Transverse cervical artery.
- b- Suprascapular artery.
- c- Ascending cervical artery.

Three veins:

- a- Internal jugular vein.
- b- Anterior jugular vein (terminal part).
- c- Subclavian vein which lies near the insertion of the muscle.

Three other structures:

- a- Phrenic nerve which crosses the muscle from lateral to medial.
- b- Deep cervical lymph nodes.
- c- Clavicle.

B- Posterior relations:

- a- Second part of subclavian artery.**
- b- Roots of brachial plexus.**
- c- Scalenus medius muscle.**

C- Medial relations:

Longus colli muscle which forms with scalenus anterior muscle an inverted V-shaped space called vertebral triangle, which contains the following

structures:

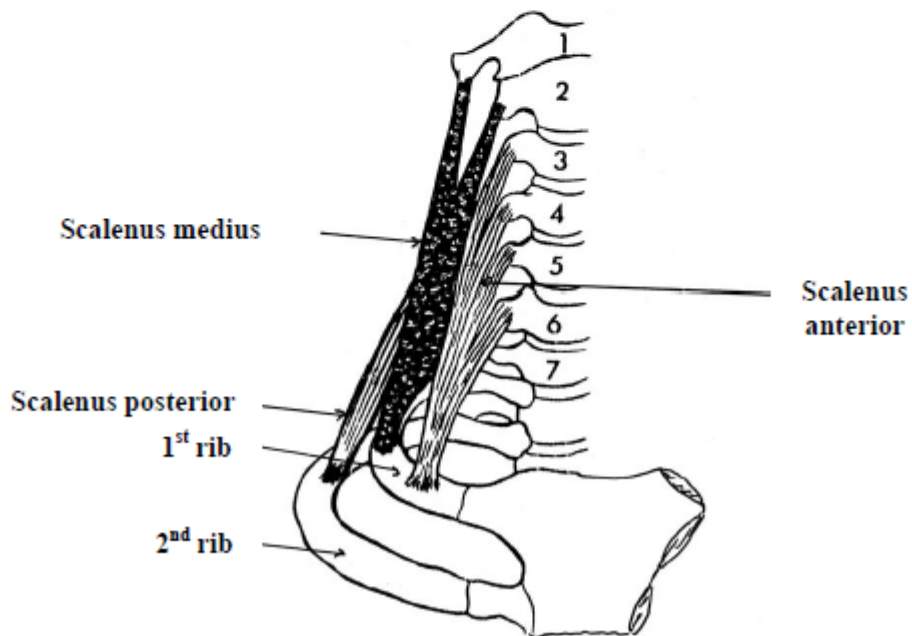
- 1. First part of subclavian artery with its branches (vertebral, thyrocervical, and internal thoracic).**
- 2. Vertebral artery ascends to enter the foramen transversarium of C6 vertebra (at the apex of the V-shaped space).**
- 3. Inferior thyroid artery crosses the space from lateral to medial near its apex, over the vertebral artery.**
- 4. Sympathetic trunk and its inferior cervical ganglion are present posteromedial to the vertebral artery.**
- 5. 7th and 8th cervical nerves.**
- 6. Common carotid artery.**
- 7. Thoracic duct on the left side only.**

D- Lateral relations:

Third part of subclavian artery and trunks of brachial plexus.

Other Scalene Muscles

Muscle	Origin	Insertion	Nerve supply	Action
Scalenus medius. The largest and longest scalene muscle	Transverse processes of 2-7 cervical vertebrae	Upper surface of 1 st rib between its tubercle and subclavian artery groove	Lower 6 cervical nerves C.3,4,5,6,7,8	It bends the neck towards the same side
Scalenus posterior. The smallest and deepest muscle	Posterior tubercles of transverse processes of (4,5,6) cervical vertebrae	External border and surface of the 2 nd rib behind serratus anterior muscle	Lower 3 cervical nerves	It bends the lower part of the neck towards its own side



Infrahyoid Muscles

Arrangements: Two superficial and 2 deep muscles

The superficial muscles:

A- Sternohyoid (medially).

B- Superior belly of omohyoid (laterally).

The deep muscles:

C- Sternothyroid (below)

D- Thyrohyoid (above)

} the oblique line of
thyroid cartilage

Attachments:

Muscle	Origin	Insertion
A- Sternohyoid: it may be absent or double	Upper posterior surface of manubrium and posterior aspect of the medial end of clavicle	Lower border of the body of hyoid bone
B- Omohyoid: 2 bellies: 1. Inferior belly 2. Superior belly They united together at angle by intermediate tendon which is attached to the medial end of the clavicle by a fibrous sling	1. Upper scapular border near the supra-scapular notch and supra scapular ligament 2. Intermediate tendon of omohyoid muscle	1. Intermediate tendon 2. Body of hyoid bone lateral to sternohyoid muscle
C- Sternothyroid: it is shorter and wider than sternohyoid and it lies deep and partly medial to it	Posterior aspect of manubrium inferior to sternohyoid and posterior edge of 1 st costal cartilage	Oblique line of thyroid cartilage
D- Thyrohyoid	Oblique line of thyroid cartilage	Body and lower border of the greater horn of hyoid bone

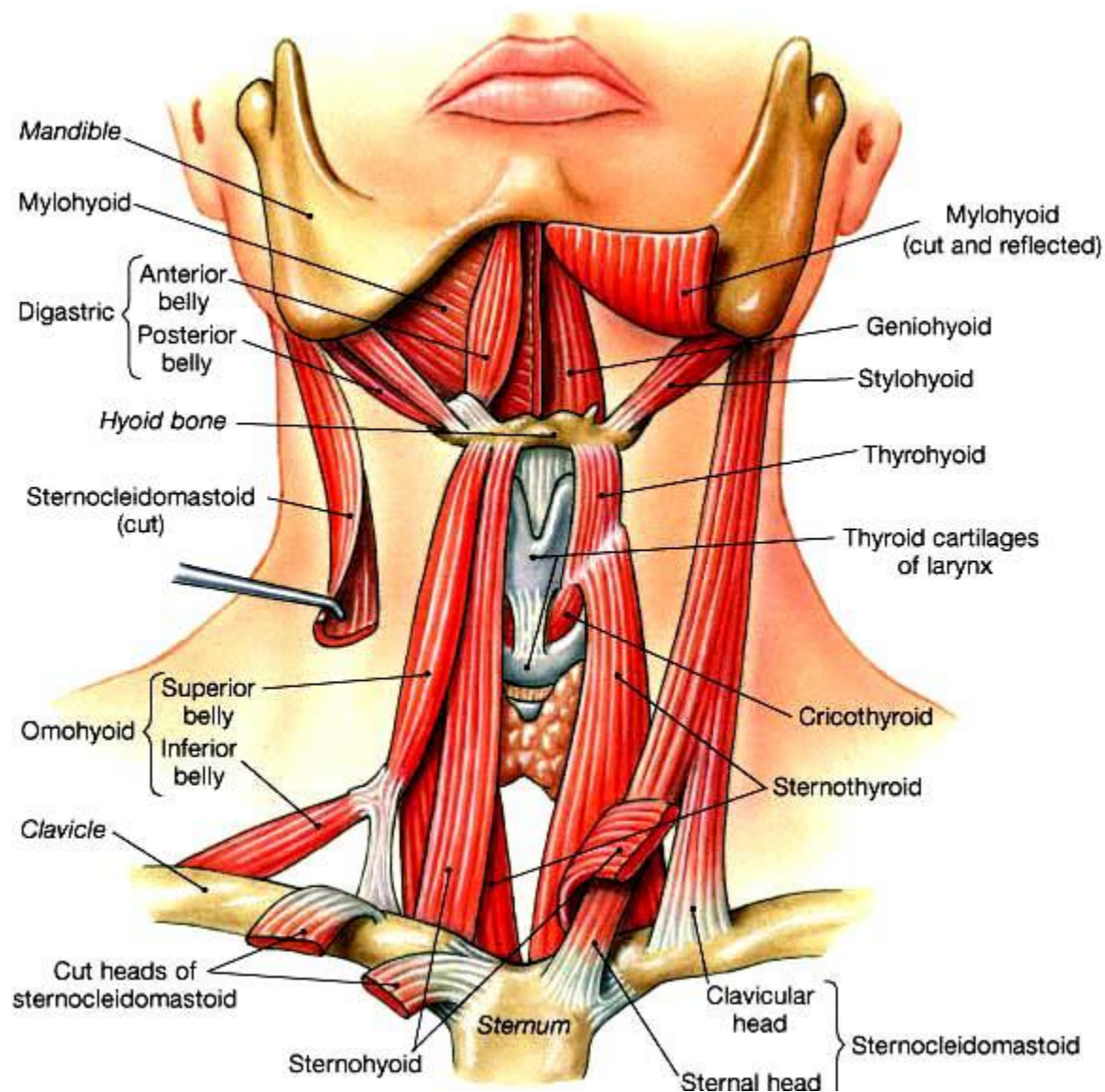
Action:

They act together during swallowing to depress hyoid bone and thyroid cartilage.

Nerve supply:

All muscles are supplied by branches from ansa cervicalis except one muscle

(thyrohyoid) which is supplied by a branch from C1 directly via hypoglossal nerve.



IV. Submental Triangle

Unpaired triangle, half of this triangle presents in each anterior triangle.

Boundaries:

- Below: Body of the hyoid bone (forms its base).
- On each side: Anterior belly of digastric.
- Above: Chin (forms its apex).

Floor: Parts of both mylohyoid muscles and median raphe.

Contents: Submental lymph nodes, and submental veins.

